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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

20CVE501 / 17CV515 – ALTERNATIVE BUILDING MATERIALS AND TECHNOLOGIES

Duration: 3 Hours

Max. Marks: 100

*Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

2) Missing data may be suitably assumed.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	Compute the embodied energy in a wall made of burnt bricks with mortar proportion 1:6, wall size is 5X3X0.23m. Take energy in burnt bricks as 4.25 MJ and energy in mortar as 1296 MJ. If this wall is built with SMB of size 230X190X100 mm using same mortar of 1:6 proportions, what is the energy saving if energy in block is 2.6MJ.	10	L*2	1	1
	b)	Explain any four environmental issues related to building materials with an example.	10	L2	2	1
2.	a)	Explain the role of construction materials and activities in global warming process.	10	L2	1	1
	b)	Explain i) cost and energy efficient roofing systems ii) efficient foundations and walls.	10	L2	2	1
3.	a)	Explain the case studies on green buildings highlighting essential principles of assessment.	10	L2	1	1,2
	b)	Explain rain water harvesting systems highlighting types and components.	10	L2	2	1
		Unit – II				
4.	a)	Explain properties and applications of any four industrial and agro wastes used as building materials.	10	L2	3	1,2
	b)	Explain matrix materials and fibers used in FRC. Give typical mix proportions of FRC.	10	L2	4	1
5.	a)	Explain the manufacturing steps of stabilized mud blocks.	10	L2	3	1,2
	b)	Explain properties and applications of any two mining and agro wastes used as building materials.	10	L2	4	1
6.	a)	Explain the properties and applications of i) Blast furnace slag ii) flyash.	10	L2	3	1
	b)	Summarise different manufacturing methods of casting ferrocement. Draw a typical cross section of ferrocement element.	10	L2	4	1
		Unit – III				
7.	a)	Explain incremental launching of bridges using sketch.	10	L2	5	1,2
	b)	Explain the concepts in Mivan formwork and its merits over conventional formwork in construction activity.	10	L2	5	1
8.	a)	Explain: i) waffle slab construction ii) modular construction methods.	10	L2	5	1,2
	b)	Outline the concept involved with filler slabs and with the help of sketch explain the casting process of Mangalore tile filler slab.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
